

Rutherford scattering 2

Find the scattering cross section for the screened Coulomb potential

$$V(r) = -\frac{Z\alpha\hbar c}{r} \exp\left(-\frac{r}{a}\right)$$

Let $f(\mathbf{p})$ be the scattering amplitude for momentum transfer vector $\mathbf{p} = \mathbf{p}_i - \mathbf{p}_f$. The following formula is the Born approximation for $f(\mathbf{p})$.

$$f(\mathbf{p}) = \frac{m}{2\pi\hbar^2} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{r}}{\hbar}\right) V(\mathbf{r}) d\mathbf{r}$$

Convert to polar coordinates where $p = |\mathbf{p}|$.

$$f(\mathbf{p}) = \frac{m}{2\pi\hbar^2} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp\left(\frac{ipr \cos \theta}{\hbar}\right) V(r, \theta, \phi) r^2 \sin \theta dr d\theta d\phi$$

Substitute the screened Coulomb potential

$$V(r, \theta, \phi) = -\frac{Z\alpha\hbar c}{r} \exp\left(-\frac{r}{a}\right)$$

to obtain

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{2\pi\hbar} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp\left(\frac{ipr \cos \theta}{\hbar} - \frac{r}{a}\right) r \sin \theta dr d\theta d\phi$$

Integrate over ϕ (multiply by 2π).

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \int_0^\pi \exp\left(\frac{ipr \cos \theta}{\hbar} - \frac{r}{a}\right) r \sin \theta dr d\theta$$

Transform the integral over θ into an integral over u where $u = \cos \theta$ and $du = -\sin \theta d\theta$. The minus sign in du is canceled by interchanging integration limits $\cos 0 = 1$ and $\cos \pi = -1$.

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \int_{-1}^1 \exp\left(\frac{ipru}{\hbar} - \frac{r}{a}\right) r dr du$$

Solve the integral over u .

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \left[\frac{\hbar}{ipr} \exp\left(\frac{ipru}{\hbar} - \frac{r}{a}\right) \right]_{u=-1}^{u=1} r dr$$

Cancel r and evaluate the limits.

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \frac{\hbar}{ip} \left[\exp\left(\frac{ipr}{\hbar} - \frac{r}{a}\right) - \exp\left(-\frac{ipr}{\hbar} - \frac{r}{a}\right) \right] dr$$

Solve the integral over r .

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \frac{\hbar}{ip} \left[\frac{1}{ip/\hbar - 1/a} \exp\left(\frac{ipr}{\hbar} - \frac{r}{a}\right) + \frac{1}{ip/\hbar + 1/a} \exp\left(-\frac{ipr}{\hbar} - \frac{r}{a}\right) \right]_{r=0}^{r=\infty}$$

Evaluate the limits. The exponentials vanish at the upper limit.

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \frac{\hbar}{ip} \left[-\frac{1}{ip/\hbar - 1/a} - \frac{1}{ip/\hbar + 1/a} \right] \quad (1)$$

Hence

$$f(\mathbf{p}) = -\frac{2mZ\alpha\hbar c}{p^2 + (\hbar/a)^2}$$

Note that $\mathbf{p}$ is momentum transfer such that

$$p^2 = |\mathbf{p}|^2 = 4mE(\cos \theta - 1)$$

Hence

$$f(\theta) = -\frac{2mZ\alpha\hbar c}{4mE(\cos \theta - 1) + (\hbar/a)^2} \quad (2)$$

Cancel m in the numerator.

$$f(\theta) = -\frac{2Z\alpha\hbar c}{4E(\cos \theta - 1) + \frac{1}{m}(\hbar/a)^2}$$

The cross section is

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2 = \left[\frac{2Z\alpha\hbar c}{4E(\cos \theta - 1) + \frac{1}{m}(\hbar/a)^2} \right]^2$$

Let $a \rightarrow \infty$ to obtain the ordinary Rutherford cross section.

$$\frac{d\sigma}{d\Omega} = \left[\frac{Z\alpha\hbar c}{2E(\cos \theta - 1)} \right]^2$$

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